OFFICIAL STAR 2.7.11b PINNED CONTROL

BLACKSTAR 1.1.0 HARDENING SOURCE blackstar.2 stack plus Q02 corrections

Q02 PUBLIC WORKLOAD MATRIX 96 threads; local storage; balanced pairs

FRAGMENTED AND SPECIALIZED BULK

Paired 76-base 20.05% measured gain 5/5 outputs exact

Paired 150-base 12.67% measured gain 3/3 outputs exact

Two-pass 20.25% measured gain 3/3 outputs exact

BySJout 31.45% measured gain 3/3 outputs exact

Chimeric 31.76% measured gain 3/3 outputs exact

Sorted BAM 18.42% measured gain 5/5 canonical BAM exact

OTHER COMMON MODES

TranscriptomeSAM 1.55% point estimate accepted noninferiority

STARsolo 10x v3 3.78% point estimate accepted noninferiority

STARlong direct RNA 19.48% measured gain 3/3 canonical SAM exact

Single-end 150-base 1.67% point estimate CV gate failed; no speed claim

36/36 MODE-SPECIFIC PAIR COMPARISONS PASSED

LIMITS one CPU family; local storage; 96-thread primary series STARlong requires symmetric seed-limit override single-end timing remains unresolved

MIXED PERFORMANCE DECISION compatibility source accepted in 1.1.0 single-end speed claim rejected